

ET 8202:

INTRODUCTION TO IS

SECURITY

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LECTURE 02

SECURITY MECHANISM

Date: 11/11/2023

SECURITY MECHANISM

- **Prevention:** Access control
- **Detection:** Auditing and intrusion detection
- **Tolerance:** Practicality

SECURITY MECHANISM

- Security mechanisms implement functions that help *prevent, detect, and respond* to security attacks
- Prevention is more fundamental
 - ✓ Detection requires that the audit trail be protected from alteration
- Sometime detection is the only option, e.g.,
 - ✓ Accountability in proper use of authorized privileges
 - ✓ Modification of messages in a network

Cryptography underlies (almost) all security mechanisms

METHODS OF DEFENCE

(Controls)

- Encryption
- Software Controls (access limitations in a data base, in operating system protect each user from other users)
- Hardware Controls (e.g. firewalls, smartcard)
- Policies (e.g.. frequent changes of passwords)
- Physical Controls (e.g. locked doors, guards)

CRYPTOGRAPHY

INTRODUCTION

What if you send any information to someone and altered information is received at the other end and causes a negative impact?

You never know how many eyes are watching you.

CRYPTOGRAPHY

INTRODUCTION

- Is there any means for securing your information or your communication?

Yes, there is.

- **CRYPTOGRAPHY is the solution to your problems.**

CRYPTOGRAPHY CONCEPTS

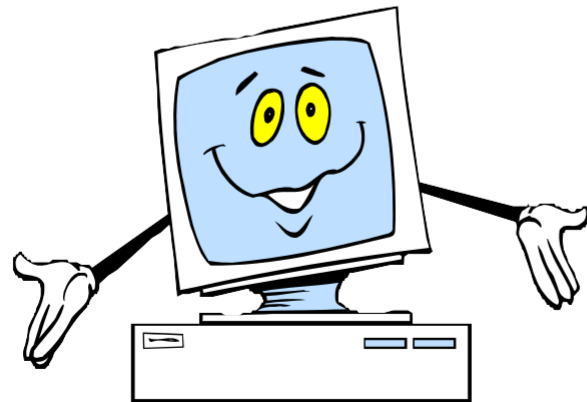


CRYPTOGRAPHY

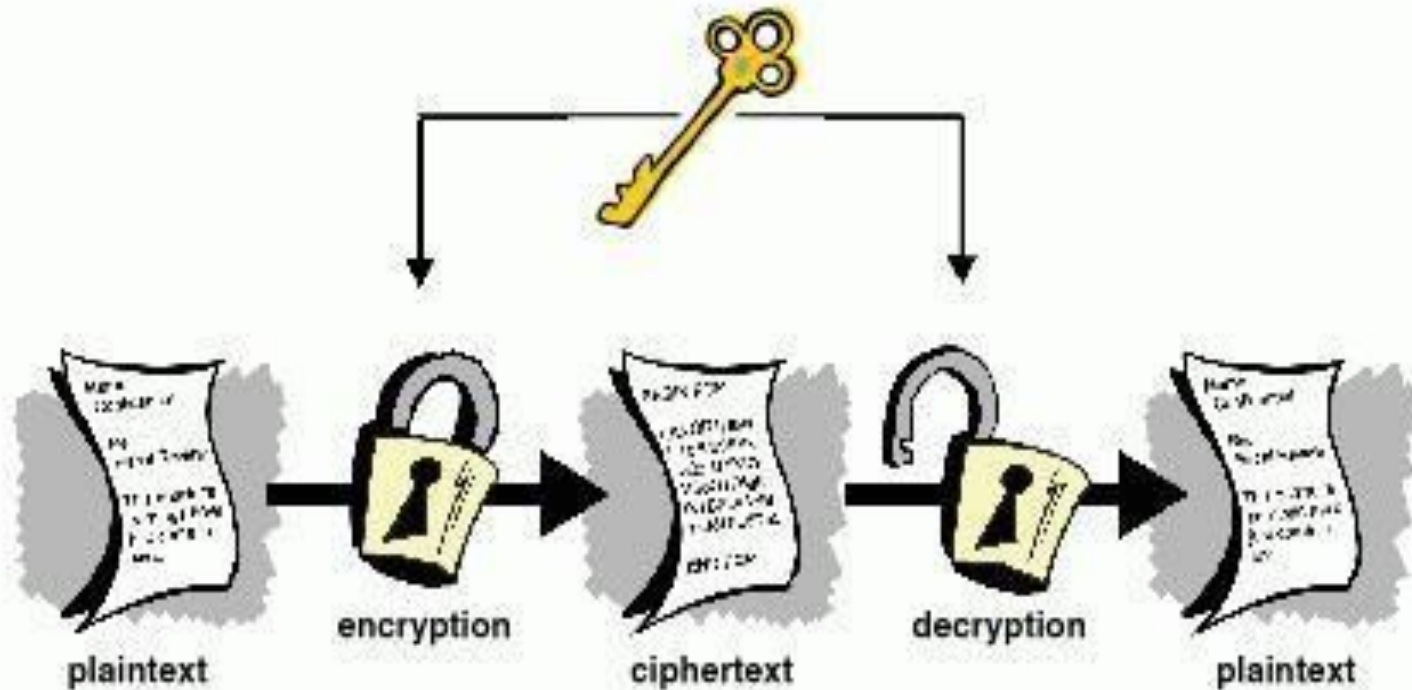
- Cryptography is the **conversion of data** into a scrambled code and send across a private or public network.
 - ✓ Original meaning: The art of secret writing.
 - ✓ Becoming a science that relies on mathematics (number theory, algebra)
 - The science of **encrypting**.
 - The science and art of designing **ciphers**.

CRYPTOGRAPHY

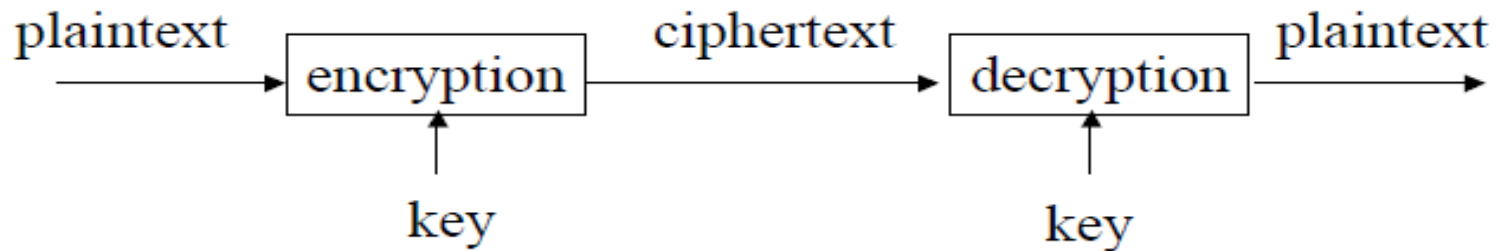
- The purpose of cryptography is to **protect** data transmitted in the likely presence of an opponent.
 - ✓ **Protect** e-mail messages, credit card information, and corporate data.
- **Objectives of cryptography**
 1. Confidentiality
 2. Integrity
 3. Authentication
 4. Non-repudiation



CRYPTOGRAPHY



TERMINOLOGIES



- **Plaintext:** A message in its original form.
 - ✓ A **message** in cryptography can be any kind of data, such as images, audio, video, text, databases, files, or data streams.
- **Ciphertext:** A message in the transformed, unrecognized form. A scrambled message. It depends on the plaintext and the secret key.
 - ✓ Designed to protect the information from an unauthorized access.

TERMINOLOGIES

- **Encryption:** The process that transforms a plaintext under the control of the **key** into a ciphertext; also known as **encode** and **encipher**.
- **Decryption:** The process that transforms an encrypted message (ciphertext) to the corresponding plaintext; also known as **decode** and **decipher**.
- **Key:** The value used to control encryption/decryption
- **Cipher** are algorithms used to encrypt or decrypt the data.
- **Cryptosystem:** a system for encryption and decryption

TERMINOLOGIES

- **Cryptanalysis (technically):** Is a study of ciphers, ciphertext or cryptosystems and to find weakness in the encryption key so that message can be decrypted without knowing the key.
 - ✓ The science of **decrypting** messages or breaking codes and ciphers.
- **Cryptanalysis (In non-technical terms):** It is an unauthorized method of recovering the original message or breaking the message. It is a combination of science, art and luck used to break messages or the entire systems.
- **Cryptology:** is a study of both cryptography and cryptanalysis.

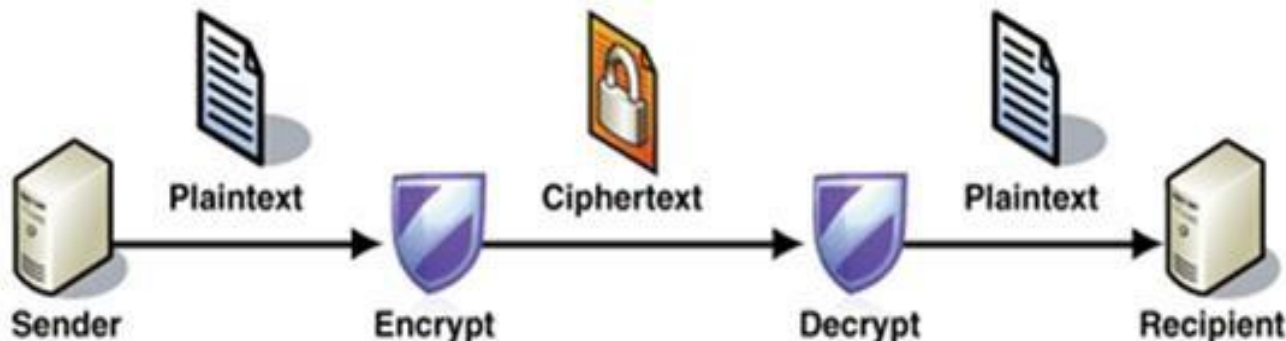
PLAINTEXT VS. CIPHERTEXT

● Basic operations

- ✓ Plaintext (P) to Ciphertext (C): **Encryption (E)**: $C = E(P)$
- ✓ Ciphertext (C) to Plaintext (P): **Decryption (D)**: $P = D(C)$
 $P = D(E(P))$

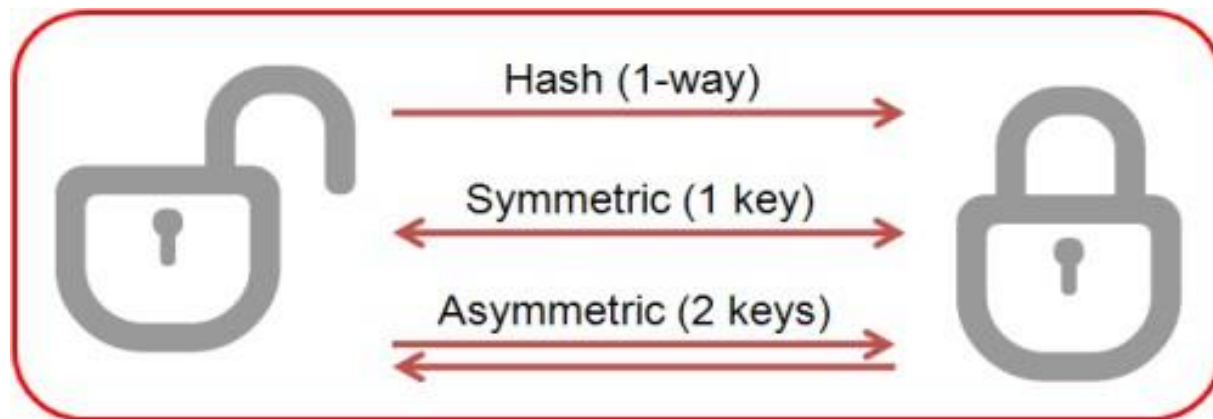
● Encryption/ Decryption with key

- ✓ Encryption key: K_E $C = E(K_E, P)$
- ✓ Decryption key: K_D $P = D(K_D, E(K_E, P))$



TYPES OF CRYPTOGRAPHY

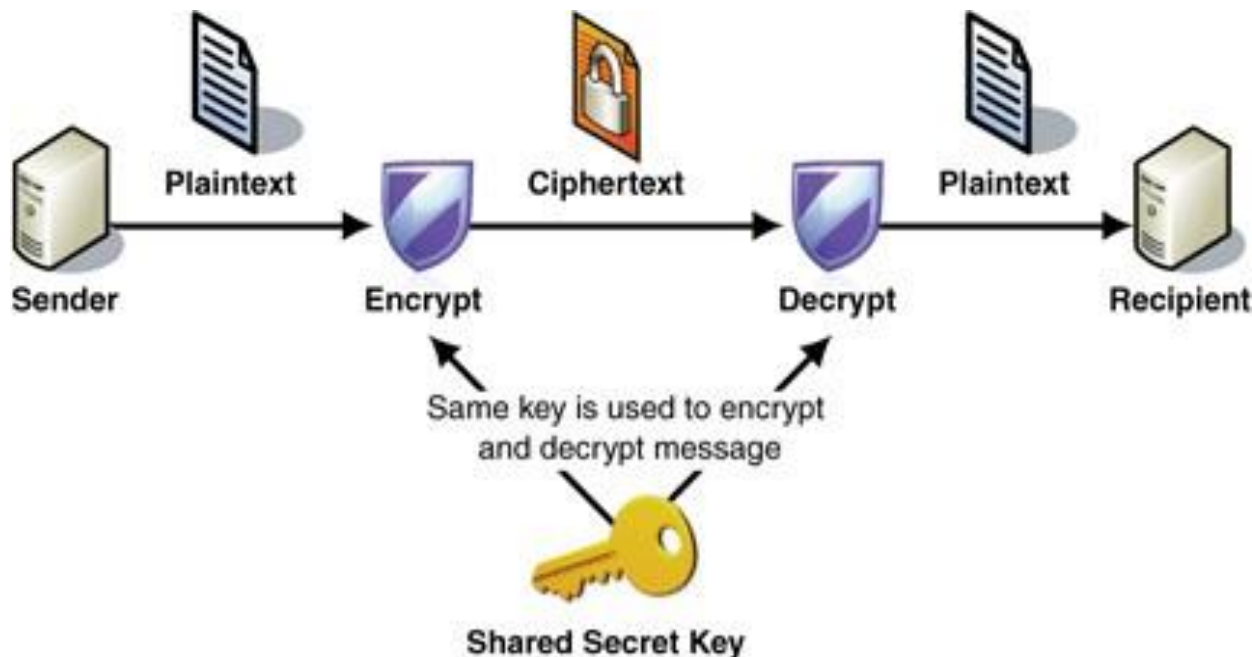
- There are many varieties of encryption with different purposes. Encryption can keep information secret, but is also used for verification.
1. **Symmetric encryption** (uses one secret key)
 2. **Asymmetric encryption** (uses a private key and a public key)
 3. **Hash function** (One-way encryption)



TYPES OF CRYPTOGRAPHY

- **Symmetric encryption**

- ✓ Symmetric encryption (secret-key, shared-key, and private-key) uses the same key for encryption as they do for decryption.



PRIVATE-KEY CRYPTOGRAPHY

- **Symmetric encryption** is a form of cryptosystem in which encryption and decryption are performed using the same key (**private/secret/single key**). It is also known as conventional encryption
- If the key is disclosed communications are compromised.
- Also known as **symmetric**, both parties are equal
 - ✓ Hence does not protect sender from receiver forging a message & claiming is sent by sender

CONVENTIONAL ENCRYPTION PRINCIPLES

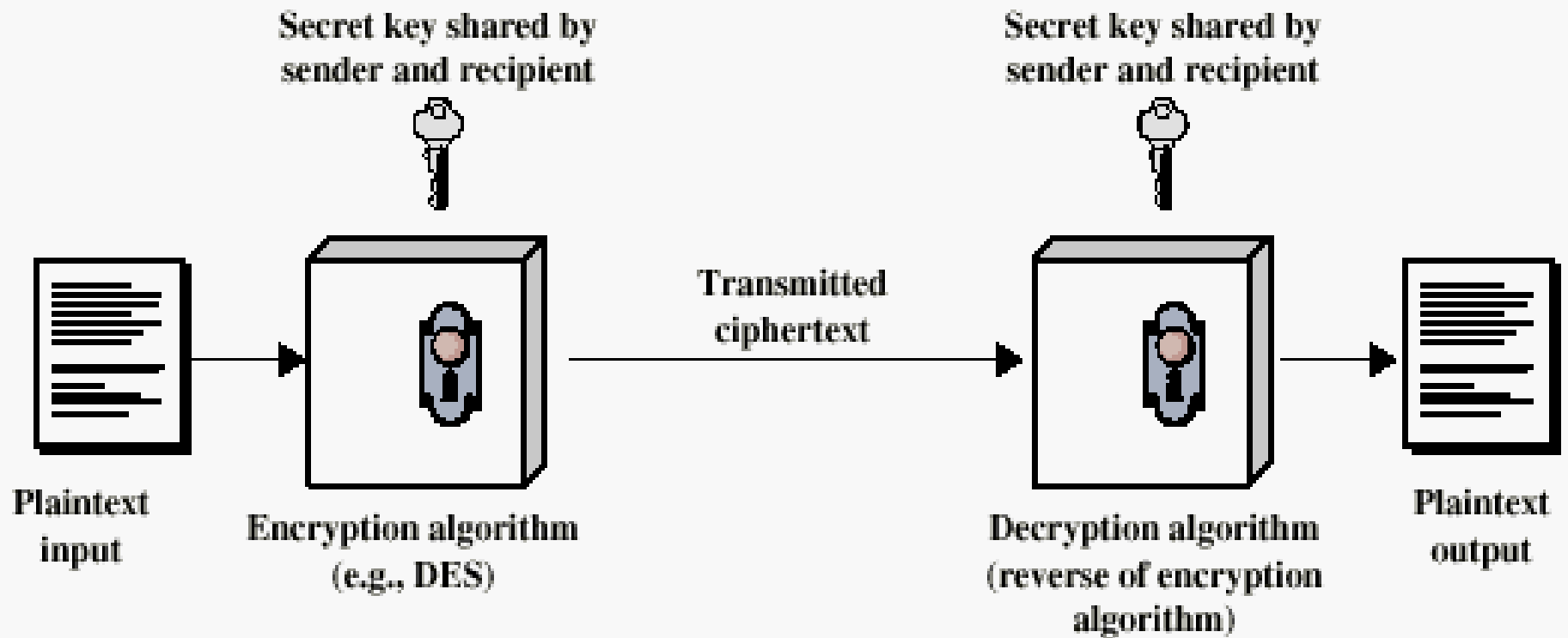


Figure 2.1 Simplified Model of Conventional Encryption

CONVENTIONAL ENCRYPTION PRINCIPLES

- An encryption scheme has five ingredients:
 - ✓ **Plaintext**
 - ✓ **Encryption algorithm**
 - ✓ **Secret Key**
 - ✓ **Ciphertext**
 - ✓ **Decryption algorithm**
- **Note:** Security depends on the secrecy of the key, not the secrecy of the algorithm

CONVENTIONAL ENCRYPTION PRINCIPLES

- An encryption scheme has **five ingredients**:
 1. **Plaintext**: This is the original intelligible message or data. Input to encryption algorithm.
 2. **Encryption algorithm**: Performs various substitutions and transformations on the plaintext. It takes the plaintext and the secret key and produces the ciphertext.
 3. **Private or Secret key**: Piece of secret data used to control encryption and decryption process.

CONVENTIONAL ENCRYPTION PRINCIPLES

4. **Ciphertext:** This is the scrambled/transformed message; An encrypted message. It depends on the plaintext and the secret key. For a given message, two different keys will produce two different ciphertexts.
5. **Decryption algorithm:** This is essentially the encryption algorithm run in reverse. It takes the ciphertext and the secret key and produces the original plaintext.

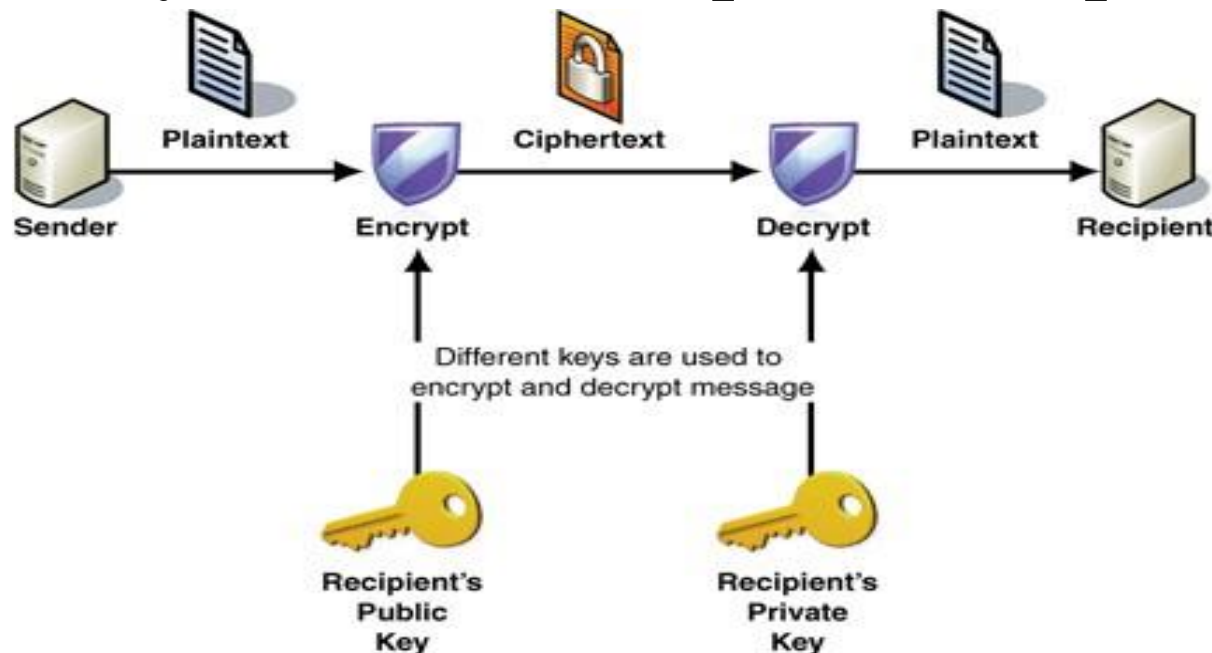
INSUFFICIENCIES WITH PRIVATE-KEY CRYPTOGRAPHY (WHY PUBLIC-KEY CRYPTOGRAPHY?)

- Developed to address two key issues:
 - ✓ **Key distribution** – how to have secure communications in general without having to trust a Key Distribution Centre (KDC) with your key
 - ✓ **Digital signatures** – how to verify a message comes intact from the claimed sender

TYPES OF CRYPTOGRAPHY

- **Asymmetric encryption**

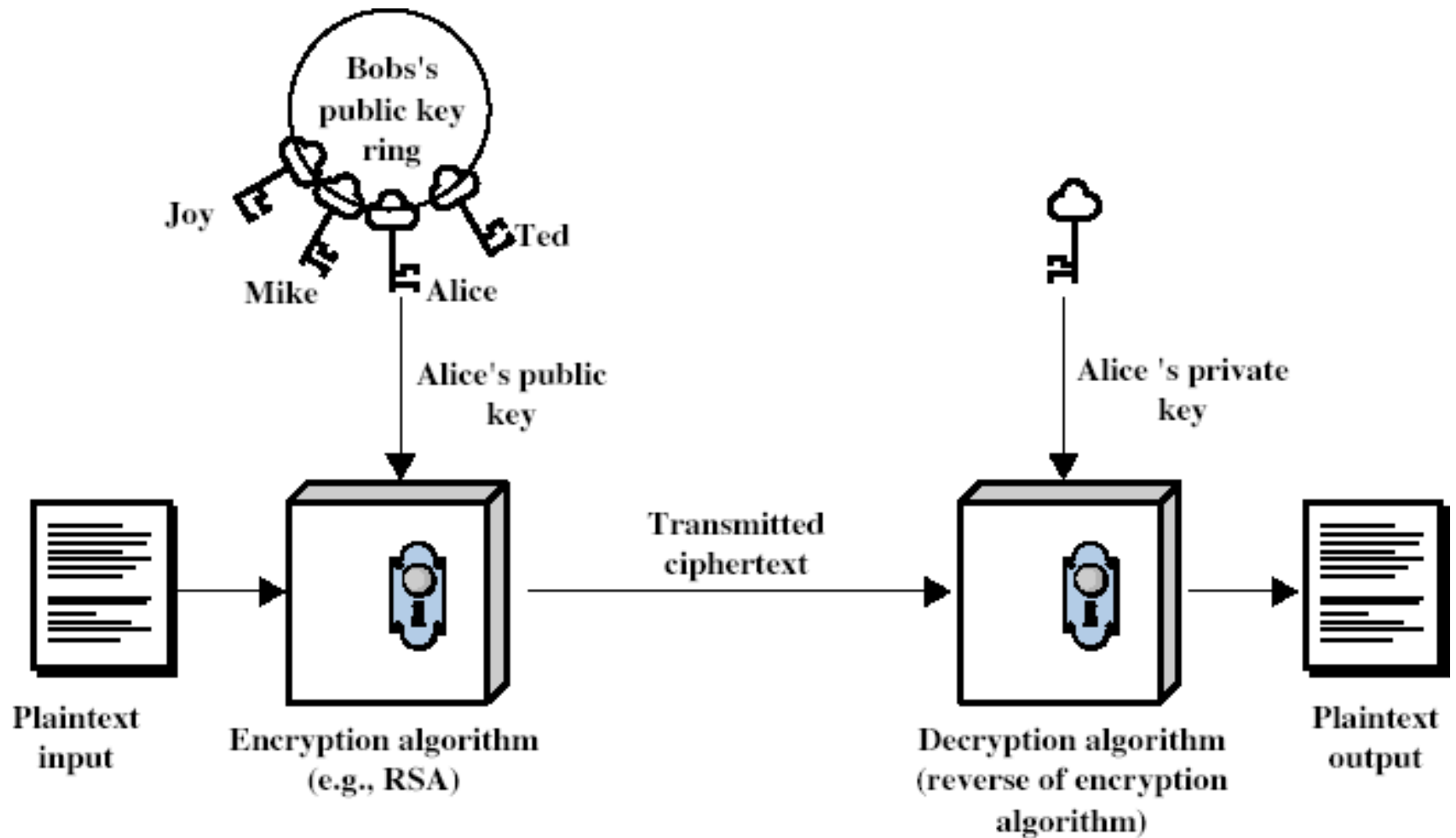
- ✓ Asymmetric encryption (public-key) uses different encryption keys for encryption and decryption. These keys are known as public and private keys.



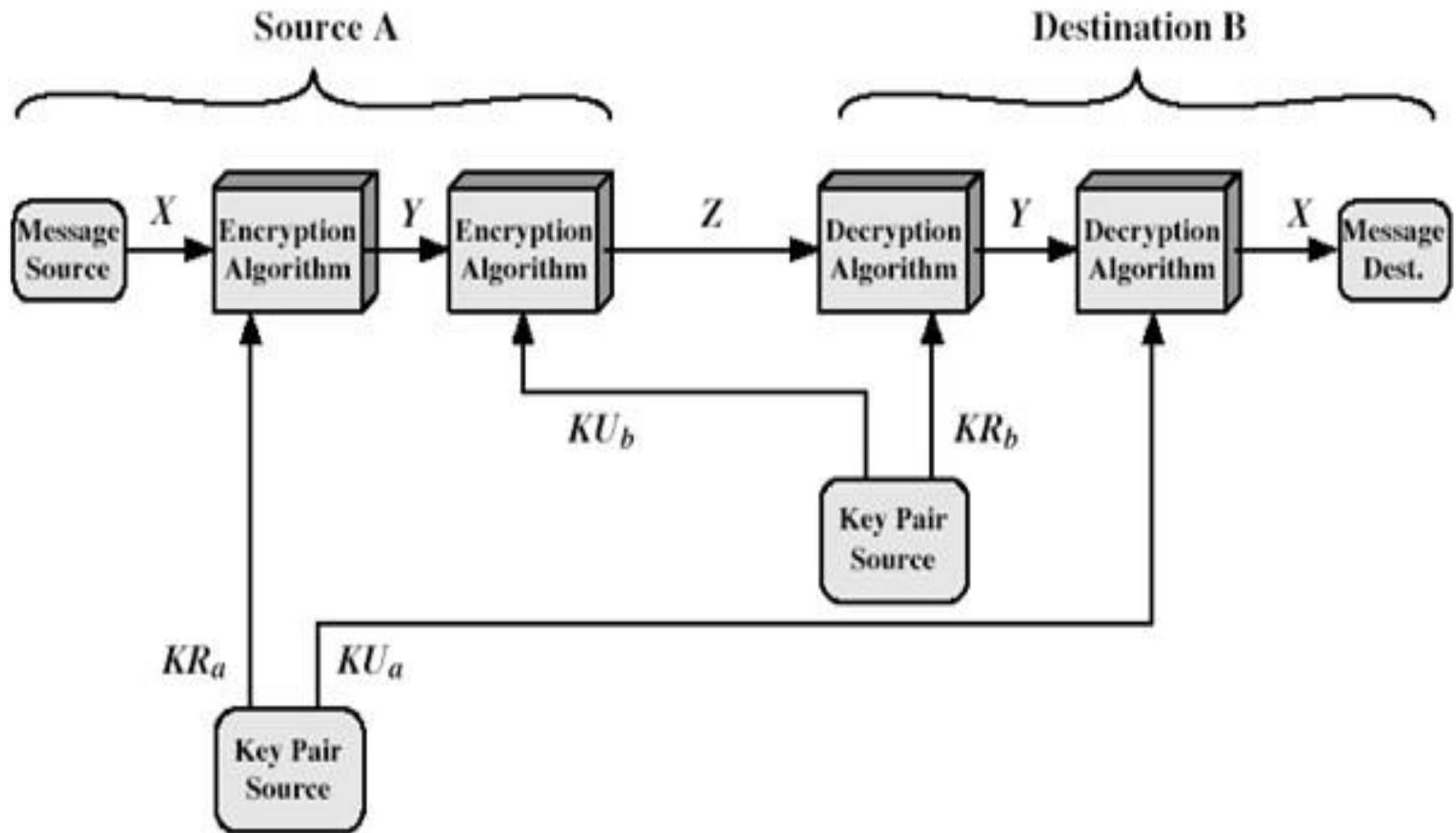
PUBLIC-KEY CRYPTOGRAPHY

- **Public-key/two-key/asymmetric** cryptography involves the use of **two** keys:
 1. **Public-key**, which may be known by anybody, and can be used to **encrypt messages**, and **verify signatures**
 2. **Private-key**, known only to the recipient, used to **decrypt messages**, and **sign** (create) **signatures**
- **Is asymmetric** because
 - ✓ Those who encrypt messages or verify signatures **cannot** decrypt messages or create signatures
- Complements **rather than** replaces private key cryptography

PUBLIC-KEY CRYPTOGRAPHY



PUBLIC-KEY CRYPTOSYSTEMS



KR-Private Key : Create Signatures, Decrypt Messages

KU-Public Key: Encrypt Messages, Verify Signature

PUBLIC-KEY APPLICATIONS

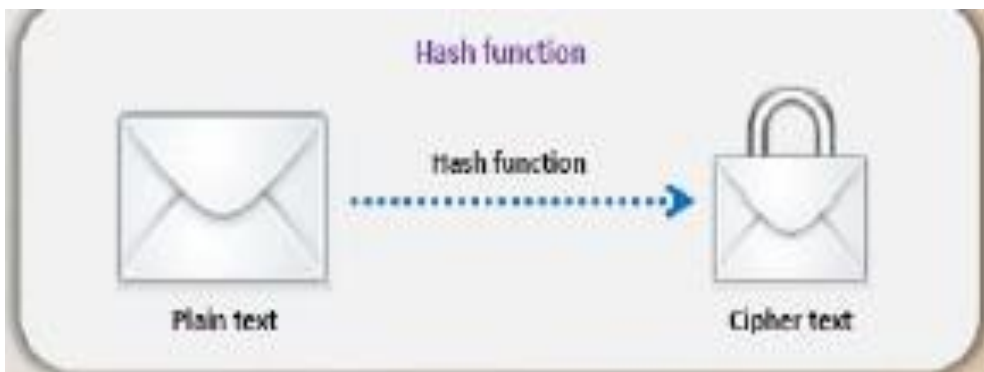
- **Can classify uses into 3 categories:**

- ✓ **Encryption/decryption** (provide secrecy) The sender encrypt message with recipient's public key
- ✓ **Digital signatures** (provide authentication) The sender sign a message with private key signature is followed by cryptographic algorithm applied to the message
- ✓ **Key exchange** (of session keys) Two side cooperation to exchange session key

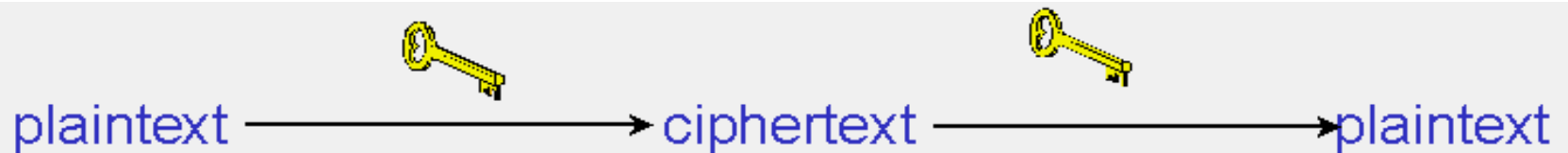
TYPES OF CRYPTOGRAPHY

● Hash function

- ✓ Hash function (message digests or one – way encryption) uses no key for encryption and decryption



THE LEXICON OF CRYPTOGRAPHY



A) Secret key (symmetric) cryptography. SKC uses a single key for both encryption and decryption.



B) Public key (asymmetric) cryptography. PKC uses two keys, one for encryption and the other for decryption.



C) Hash function (one-way cryptography). Hash functions have no key since the plaintext is not recoverable from the ciphertext.

CRYPTOGRAPHY

Cryptographic systems are **characterized** along three independent dimensions



1. The type of **operations** used for transforming plaintext to ciphertext.
2. The number of **keys** used.
3. The way in which the plaintext is **processed**.

CRYPTOGRAPHY

- Cryptographic systems are classified along three independent dimensions:
 1. The type of **operations** used for transforming plaintext to ciphertext. Based on two principles (substitution & transposition).
 2. The number of **keys** used
 - Symmetric or conventional encryption (single key)
 - Asymmetric (two-keys, or public-key encryption)
 3. The way in which the plaintext is processed
 - **Block cipher:** Encrypts block of data of fixed size.
 - **Stream cipher:** Encrypts continuous streams of data (one bit or one byte at a time).

(1) The type of operations used for transforming plaintext to ciphertext

- All encryption algorithms are based on two general principles:
 - ✓ **Substitution cipher:** It replaces bits, characters, or blocks of characters with different bits, characters or blocks. e.g. Caesar cipher, hill cipher.
 - ✓ **Transposition cipher:** The letters of the plaintext are shifted about to form the cryptogram. e.g. Rail fence.

(2) The number of keys used.

- General there are two types of encryption :
 1. **Symmetric, single-key, secret-key, or conventional encryption.**
 - ✓ If both sender and receiver use the same key.
 - ✓ Same key is used for encryption and decryption
 2. **Asymmetric, two-key, or public-key encryption.**
 - ✓ If the sender and receiver use different keys.
 - ✓ Different keys are used for encryption and decryption

(3) The way in which the plaintext is processed

- **Two types**

1. **Block cipher:** Encrypts block of data of fixed size.
2. **Stream cipher:** Encrypts continuous streams of data (one bit or one byte at a time).

CRYPTANALYSIS

- The objective of attacking encryption system is to recover the key in use rather than simply to recover the plaintext of a single ciphertext. Two general approaches:

- ✓ **Cryptanalysis**

- This type of attack exploits the characteristics of algorithm to attempt to deduce a specific plaintext or to deduce the key being used.

- ✓ **Brute-Force Attack**

- The attacker tries every possible key on a piece of ciphertext until intelligible translation into plain text is obtained.

STRONG CRYPTOGRAPHY

- The opponent should be unable to decrypt ciphertext or discover the key even if he or she is in possession of a number of **ciphertext** with the **plaintext** that produced each cipher text.
- Sender and receiver must have obtained copies of the secret key in a secure fashion and must keep the key secure.

ASSUMPTION

- We do not need to keep the algorithm secret, we need to keep only the key secret.
- We assume that it is impractical to decrypt a message on the basis of the cipher text plus knowledge of encryption/decryption algorithm.

STRONG CRYPTOGRAPHY

- An encryption scheme is **unconditionally secure** if
 - ✓ The ciphertext generated by the scheme does not contain enough information to determine uniquely the corresponding plaintext, no matter how much ciphertext is available.
- An encryption scheme is said to be **computationally secure** if:
 1. The cost of breaking the cipher exceeds the value of the encrypted information, and
 2. The time required to break the cipher exceeds the useful lifetime of the information.

TYPES OF ATTACKS ON ENCRYPTED MESSAGES

● **Ciphertext only**

- ✓ In this attack the attacker knows only the ciphertext to be decoded. The attacker will try to find the key or decrypt one or more pieces of ciphertext (only relatively weak algorithms fail to withstand a ciphertext-only attack).

● **Known plaintext**

- ✓ The attacker has a collection of plaintext-ciphertext pairs and is trying to find the key or to decrypt some other ciphertext that has been encrypted with the same key.

TYPES OF ATTACKS ON ENCRYPTED MESSAGES

- **Chosen plaintext**

- ✓ This is a known plaintext attack in which the attacker can choose the plaintext to be encrypted and read the corresponding ciphertext.

- **Chosen ciphertext**

- ✓ The attacker has the able to select any ciphertext and study the plaintext produced by decrypting them.

CLASS ACTIVITY

1. What is the effect if either of the attack succeeds in deducing the key?
2. Why ciphertext-only attack is the easiest to defend against hacker?

ENCRYPTION ALGORITHMS

CIPHER

- **Cipher** are algorithms used to encrypt or decrypt the data.

Two Types

1. Classical Ciphers

- Substitution cipher
- Transposition cipher

2. Modern Cipher

- Based on types of key used
 - ✓ Private key ✓ Public key
- Based on the types of input data
 - ✓ Block cipher ✓ Stream cipher

SUBSTITUTION CIPHER

SUBSTITUTION TECHNIQUES

- A **monoalphabetic substitution cipher** maps a plaintext alphabet to a ciphertext alphabet, so that each letter of the plaintext alphabet maps to a single unique letter of the ciphertext alphabet.
- The **Caesar cipher** involves replacing each letter of the alphabet with the letter standing k places further down the alphabet, for k in the range 1 through 25.

KEY POINTS

- Plaintext is always in **lowercase**
- Ciphertext is in **uppercase**
- Key values are ***italicized lowercase***

CAESAR CIPHER

- **Original meaning:** The Caesar cipher involves replacing each letter of alphabet with the letter standing three places further down the alphabet.
- **General Caesar algorithm:** The Caesar cipher involves replacing each letter of the alphabet with the letter standing k places further down the alphabet, for k in the range 1 through 25.

Plain	meet	me	after	the	toga	party
Cipher	PHHW	PH	DIWHU	WKH	WRJD	SDUWB

plain:	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
CIPHER:	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

CAESAR CIPHER

- Let us assign numerical equivalent to each letter:

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

- The algorithm can be expressed as follows:

$$C = E(3, p) = (p + 3) \bmod 26 \text{ \{Arithmetic Modular\}}$$

- The shift may be of any amount, so that the general Caesar algorithm:

$$C = E(k, p) = (p + k) \bmod 26$$

Where k takes the value in the range 1 to 25.

The decryption algorithm is simple:

$$p = D(k, C) = (C - k) \bmod 26$$

BRUTE-FORCE CRYPTANALYSIS

- Simply try all the 25 possible keys:
 - ✓ Assumptions:
 - The encryption and decryption algorithms are known;
 - There are only 25 keys to try;
 - The language of the plaintext is known and easily recognizable.

EXAMPLE OF BRUTE-FORCE CRYPTANALYSIS

	PHHW	PH	DIWHU	WKH	WRJD	SDUWB
Key						
1	oggy	og	chvgt	vjg	vgic	rctva
2	nffu	nf	bgufs	uif	uphb	qbauz
3	meet	me	after	the	toga	party
4	idda	id	zesdq	sgd	snfz	ozqsx
5	kccr	kc	ydrp	rfc	rmey	nyprw
6	jbbq	jb	xcqbo	qeb	qldx	mxoqv
7	iaap	ia	wbpan	pda	pkcw	lwnpu
8	hzzo	hz	vaozm	ocz	objv	kvmot
9	gyyn	gy	uznyl	nby	niau	juins
10	fxxm	fx	tymxk	max	mhzt	itkmr
11	ewwl	ew	sxlwj	lzw	lgys	hsjlg
12	dvvk	dv	rwkvi	kyv	kfxr	qrikp
13	cuuj	cu	qvjuh	jxu	jweq	fghjo
14	assh	as	othsf	hvs	hcuo	dofhm
15	btti	bt	pultg	iwt	idvp	epgin
16	zrrg	zr	nsrge	gur	gbtn	cnegl
17	yggf	yg	mrfgd	ftq	faam	bmdfk
18	xppe	xp	lqepc	esp	ezrl	alcej
19	wood	wo	kpdob	dro	dyqk	zkbdi
20	vnnb	vn	jocna	cqn	cxpj	yjach
21	ummb	um	inbmz	bpm	bwol	xizbg
22	tlla	tl	hmaly	aol	avnh	whyaf
23	skkz	sk	qlzcx	znk	zumg	vgxze
24	rjjy	rj	fkyjw	ymj	ytlf	ufwyd
25	qiix	qi	ejxiv	xli	xske	tevx

CAESAR CIPHER

- Advantages and Disadvantages of the Caesar Cipher
- **Advantage:** Easy to use
- **Disadvantage**
 - ✓ The only problem with this cryptosystem is that it is easily broken. That is, it is possible for unauthorized person to convert the ciphertext back to plaintext.

MONOALPHABETIC CIPHER

- A dramatic increase in the key space can be achieved by allowing an arbitrary substitution.

plain:	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
CIPHER:	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

- If, instead, the “cipher” line can be any permutation of the 26 alphabetic characters then there are $26!$ Or greater than 4×10^{26} possible keys. This is 10 orders of magnitude greater than the key space for DES and would seem to eliminate brute-force techniques for cryptanalysis. This is referred to **“monoalphabetic substitution cipher”**

ANOTHER LINE OF ATTACK

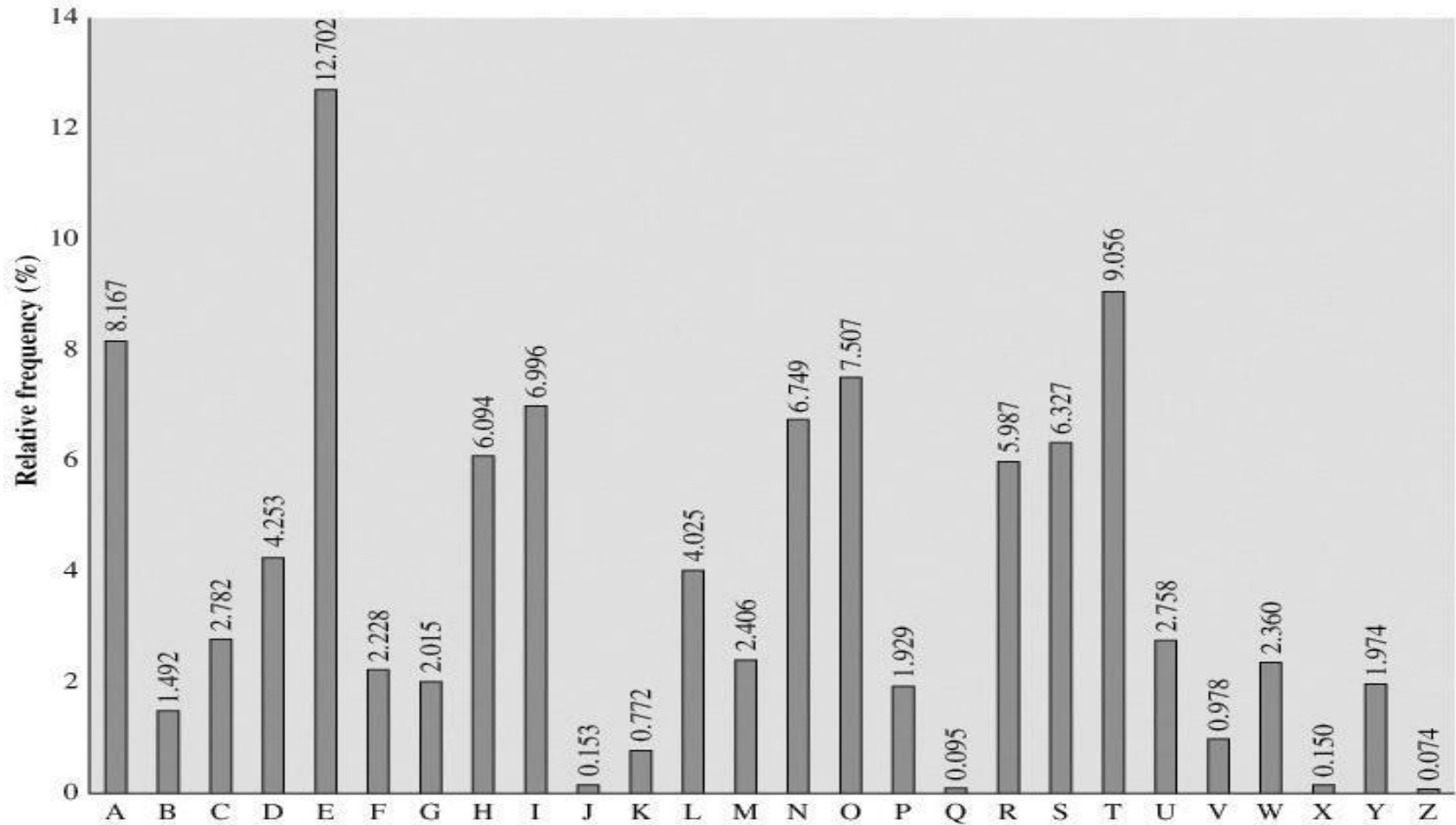
- If the cryptanalyst knows the nature of the plain text (eg. Noncompressed English text), then the analyst can exploit the regularities of the language (**Cryptanalysis using frequency table**).

The frequencies of occurrence of letters constitute an elementary characteristic of a natural language.

In English, the most frequent letters are E, T, A, O, N, R, I, S, and H.

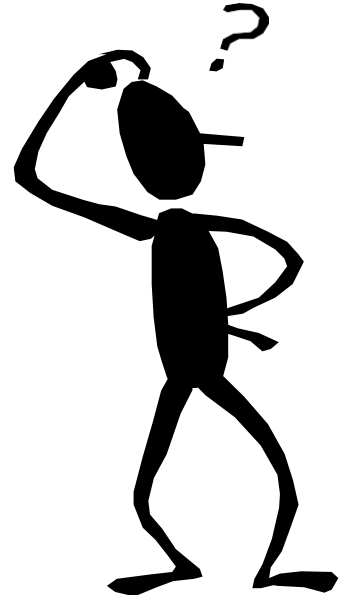
Roughly 13% of the letters in a large sample of English text should be E's.

Relative Frequency of Letters in English Text



- In English, the most frequent letters are E, T, A, O, N, R, I, S, and H.

THANK YOU



END

ET 8202: LECTURE 02

(A)